

Lab Experiment 6: Manage Users and Groups

Objective

To understand the concepts of **Users and Groups** in Linux and learn how to **create, modify, and delete** user and group accounts using basic commands.

Background

In Linux:

- Every individual who uses the system is a **user**.
- A **group** is a collection of users that helps the administrator assign permissions easily.

Instead of setting file permissions for each user one by one, Linux lets you assign them to a **group** and manage permissions collectively.

Difference Between Users and Groups

Feature	User	Group
Meaning	An individual who can log into the system	A collection of users with shared permissions
Example	student1, teacher, admin	students, faculty, developers
Purpose	Represents one person	Helps manage permissions for several users together
Stored In	/etc/passwd	/etc/group
Password	Yes (each user has their own password)	Usually no password
Home Directory	Yes (e.g., /home/student1)	No
Creation Command	sudo useradd	sudo groupadd
Modification Command	sudo usermod	sudo gpasswd
Deletion Command	sudo userdel	sudo groupdel

Steps to Perform in the Lab

Step 1: Check Current User

View which user is currently logged in:

```
whoami
```

View detailed user information:

```
id
```

```
ls -ltr
```

```
drwxr-xr-x  2 masood masood 4096 Nov 01 09:45 masood
```

Permissions (d=directory, rwx=read/write/execute)	Number of hard links	User owner	Group owner	Size in bytes	Modification date & time	File/directory name
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Step 2: Create a New User

Create a new user named student1:

```
sudo useradd -m student1
```

Here:

- `-m` → creates a home directory automatically.

(check `ls -ltr`)

Set a password for the user:

```
sudo passwd student1
```

Verify the user's creation:

```
grep student1 /etc/passwd
```

Step 3: Create a New Group

Create a group named class1:

```
sudo groupadd class1
```

Check if the group exists:

```
grep class1 /etc/group
```

Step 4: Modify a User (Using usermod)

Use the `usermod` command to **modify** user details.

1. Add a user to a group

```
sudo usermod -aG class1 student1
```

`-aG` → adds the user to a new group (without removing existing ones).

Change the group owner of student1's home directory to class1

```
sudo chgrp class1 /home/student1
```

(Now the username becomes `studentA`.) (Check using `ls -ltr`)

2. Change the username

```
sudo usermod -l studentA student1
```

(Now the username becomes `studentA`.)

3. Change the home directory of a user

```
sudo usermod -d /home/studentA -m studentA
```

4. Check which groups a user belongs to

```
groups studentA
```

Step 5: Change User Password

Change the password for a user:

```
sudo passwd studentA
```

Step 6: Delete User or Group

Delete a user (keep home directory):

```
sudo userdel studentA
```

Delete a user (with home directory):

```
sudo userdel -r studentA
```

Delete a group:

```
sudo groupdel class1
```

Check to confirm removal:

```
grep studentA /etc/passwd
grep class1 /etc/group
```

Outcome

After completing this lab, students will be able to:

- Understand the **difference between users and groups**.
- Create, modify, and delete **user accounts**.
- Create and manage **groups**.
- Use the **usermod** command to modify user details.
- Assign users to groups for shared permissions.

Viva Questions

1. What is the difference between a user and a group?
2. Which file stores all user information in Linux?
3. What does the **-m** option in **useradd** do?
4. What is the purpose of the **usermod** command?
5. How do you add a user to a group?
6. Which command changes the username of a user?
7. How do you check which groups a user belongs to?
8. How do you delete a user account?
9. Which file stores group information?
10. Why is group management useful in Linux?